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Project Title:	Epigenetic effects of radiation on epithelial cell self-renewal		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Zip Code:	94720-8099	**Congressional District:**	9
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Flight Program:			
Flight Assignment:	NOTE: Interagency agreement with DOE (per ARC)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Parvin, Bahram (Lawrence Berkeley National Laboratory) Kronenberg, Amy (Lawrence Berkeley National Laboratory)		
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Task Description:	Aberrant self-renewal is a fundamental property of cancer cells that can be conferred by non-mutational (epigenetic) means. Early events in radiation-induced carcinogenesis must either cause the expansion of pre-existing cells with extensive self-renewal potential or the acquisition of extensive self-renewal potential by cells that have repressed it. A current weakness in cancer risk assessment models is the lack of consideration of epigenetic factors that influence malignant transformation. Recent data from our own lab as well as others indicate that radiation can promote changes in the epigenetic properties of cultured human cell populations and intact rodent organs. Using a three-dimensional cell culture system that supports tissue-like behavior of human breast cells, we will: 1) determine the effects of ionization density and dose on the frequency of self-renewal versus differentiation, 2) identify biochemical pathways that show persistent changes as functions of ionization density and dose, and 3) use genetic manipulations to test the mechanistic involvement of discrete pathway components in persistent radiation quality effects. These experimental studies will employ new assays to improve the mechanistic understanding of radiation quality effects on the gain of aberrant self-renewal capacity and loss of responsiveness to differentiation signals--two related cancer development processes. The intent of this project is to develop a breast tissue-specific risk model of major processes in radiation carcinogenesis. Enhanced self-renewal and loss of differentiation are important surrogate endpoints that can be used directly in estimates of cancer risk and in reducing the uncertainties in predicting radiation quality effects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are developing assays of aberrant breast cell differentiation and self-renewal. These important surrogate endpoints will be useful in better estimating cancer risk associated with radiation and other potential carcinogens.
Task Progress:	To avoid selection bias associated with expansion in adherent cultures, we have performed irradiation experiments using thawed primary epithelial cell-enriched human breast tissue fragments (organoids) in suspension. Our general scheme has involved the following steps: 1) overnight incubation of thawed organoids in serum-free medium, 2) acute irradiation, 3) overnight recovery, 4) dissociation to single cells, 5) growth and differentiation assays. Using organoids from two individuals, we have detected dose-dependent increases in the percentages of “side-population” cells present after irradiation with up to 5 Gy of X-rays. Side population cells identified using this technique are those cells that do not retain the DNA-binding dye, Hoechst 33342, usually because they express high levels of the ATP-binding Cassette Transporter ABCG2 that actively pumps out the Hoechst dye. This assay is one of the oldest and most trusted techniques for identifying and sorting stem and early progenitor cells in a variety of tissues and species. During the spring 2011 run (NSRL11A), organoid aliquots from two specimens were exposed to 50 and 100 cGy doses of gamma rays, as well as 25 and 50 cGy doses of Si and Fe. After overnight recovery, the control and irradiated samples were dissociated to single cells, and aliquots shipped back to LBNL for side population assays. Changes in proportions of SP cells were noted, but no consistent pattern emerged. Because of uncertainties inherent in the shipment of live cells cross-country for analysis, this assay was abandoned in favor of assays that could be performed onsite at BNL, or after fixation. The first such assay involved staining and sorting for c-kit, a cell surface marker that others have shown to be highly expressed by luminal estrogen receptor negative progenitors in both mouse and human mammary glands. We found that ~ 1% of the cells within unirradiated organoids expressed c-kit, and that the level was fairly constant in irradiated samples, with the exception of one set of samples irradiated with the highest dose (0.5 Gy) of Fe ions, where the percentage of c-kit(+) cells was significantly decreased. Beginning with the fall 2011 run (NSRL11C), we examined the effects of radiation on a second set of cell surface markers, CD44 and CD24. The fraction of CD44hi/CD24Lo cells has proven useful for distinguishing stem-like cells in both malignant and non-malignant human breast tissues. To evaluate the possibility that the trypsinization procedure itself would alter the relative abundance of CD44hi /CD24Lo cells, we subjected aliquots of the same organoid preparation to trypsin for differing amounts of time. The results demonstrated that trypsinization for periods between 5 and 20 minutes did not lead to differences in the fractions of CD44hi/CD24Lo cells, confirming that the dissociation procedure itself was not a selective influence. Results obtained with organoids from several individuals, each run in triplicate, indicated that the baseline fraction of cells with this phenotype was relatively stable for each specimen, but ranged between 0.5 and 5% among specimens. During the fall 2011 run (NSRL11C), organoid aliquots from two specimens were exposed to 50 and 100 cGy of gamma rays, 25 and 50 cGy doses of Si (300 MeV/amu), as well as 10 and 25 cGy doses of Fe (300 MeV/amu and 1 GeV/amu). The results indicated that, at the low to moderate doses of radiation employed, no significant changes in the fractions of CD44hi/CD24Lo cells were detectable 24 hours after irradiation, regardless of ion or charge density. To assess the differentiation potential of primary human mammary epithelial cells (HMECs) derived from control or irradiated organoids, 24 hours after irradiation the organoids were dissociated with trypsin to single cells. The isolated cells were then incubated at clonal densities in adherent cultures for 7 days before fixation and staining with specific antibodies for keratins 14 (basal marker) and/or 19 (luminal marker). In preliminary experiments, when irradiated with 1 Gy of X-rays, the percentages of mixed colonies containing both luminal and basal cells were significantly higher for one specimen and marginally higher for another specimen, than for corresponding unirradiated controls. During the fall 2011 run (NSRL11C), organoid aliquots from three specimens were exposed to the ions and doses indicated above. Colonies were generated and stained. Preliminary results from one specimen (N158) exposed to gamma rays or 1 GeV/amu Fe have not shown statistically significant differences in the percentages of mixed colonies generated from unirradiated versus irradiated organoids. Subsequent exposure of organoids from this same specimen to 1 Gy of X-rays at LBNL also did not yield statistically significant differences in the percentages of mixed colonies. The variability observed is therefore likely to be attributable to heterogeneity among the specimens themselves, rather than to beam characteristics or uncontrolled technical issues. Colonies from additional specimens and exposures are currently being evaluated. During the fall 2011 run, we also assessed the colony-forming efficiency of individual cells from a single specimen (N158) that were irradiated as organoids, maintained as organoids for 24 hours post-irradiation, and dissociated to permit single cells to form colonies. Organoids from the same individual were exposed to gamma rays or 1 GeV/amu Fe ions. The results, to date, demonstrate that graded low doses of gamma rays do not elicit measurable toxicity in this standard assay, suggesting that when left intact, the local physical environment exerts a protective effect on potential

	clonogens for our sparsely ionizing reference radiation. In contrast, exposure to graded low doses of 1 GeV/amu Fe ions results in increasing cytotoxicity. Additional samples are being analyzed now for organoids from the same specimen exposed to 300 MeV/amu Fe ions or 300 MeV/amu Si ions. We have also acquired data using this assay for organoids from additional specimens, and the analysis of these results for all four irradiation types (gamma, Si (300 MeV/amu), Fe (300 MeV/amu), and Fe (1000 MeV/amu)) is underway.
Bibliography Type:	Description: (Last Updated: 05/28/2013)
Abstracts for Journals and Proceedings	Yaswen P, Kaur G, Gauny S, Parvin B, Kronenberg A. "Epigenetic effects of radiation on epithelial cell self-renewal." Presented at the 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7031.pdf , Sep-2011